SG No. 91 D_4^3 $P4_122$ [tetragonal]

* plus set: $+[0, 0, 0]$

* Wyckoff site: `4a`, site symmetry: `.2.`

Table 1: Wyckoff bond: `4a@4a`, bond symmetry: `.2.`

No.	vector	center	mapping
1	$[X, 0, Z]$	$[0, y, 0]$	`[1,-5]`
2	$[-X, 0, Z]$	$\left[0, -y, \frac{1}{2}\right]$	`[2,-6]`
3	$[0, X, Z]$	$\left[-y, 0, \frac{1}{4}\right]$	`[3,-8]`
4	$[0, -X, Z]$	$\left[y, 0, \frac{3}{4}\right]$	`[4,-7]`

Table 2: Wyckoff bond: `4b@4a`, bond symmetry: `.2.`

No.	vector	center	mapping
1	$[0, X, 0]$	$[0, y, 0]$	`[1,5]`
2	$[0, -X, 0]$	$\left[0, -y, \frac{1}{2}\right]$	`[2,6]`
3	$[-X, 0, 0]$	$\left[-y, 0, \frac{1}{4}\right]$	`[3,8]`
4	$[X, 0, 0]$	$\left[y, 0, \frac{3}{4}\right]$	`[4,7]`

Table 3: Wyckoff bond: `8c@4a`, bond symmetry: `1`

No.	vector	center	mapping
1	$[X, Y, Z]$	$[0, y, 0]$	`[1]`
2	$[-X, -Y, Z]$	$\left[0, -y, \frac{1}{2}\right]$	`[2]`
3	$[-Y, X, Z]$	$\left[-y, 0, \frac{1}{4}\right]$	`[3]`
4	$[Y, -X, Z]$	$\left[y, 0, \frac{3}{4}\right]$	`[4]`
5	$[-X, Y, -Z]$	$[0, y, 0]$	`[5]`
6	$[X, -Y, -Z]$	$\left[0, -y, \frac{1}{2}\right]$	`[6]`
7	$[Y, X, -Z]$	$\left[y, 0, \frac{3}{4}\right]$	`[7]`
8	$[-Y, -X, -Z]$	$\left[-y, 0, \frac{1}{4}\right]$	`[8]`

* Wyckoff site: `4b`, site symmetry: `.2.`

Table 4: Wyckoff bond: `4a@4b`, bond symmetry: `.2.`

No.	vector	center	mapping
1	$[X, 0, Z]$	$\left[\frac{1}{2}, y, 0\right]$	`[1,-5]`
2	$[-X, 0, Z]$	$\left[\frac{1}{2}, -y, \frac{1}{2}\right]$	`[2,-6]`
3	$[0, X, Z]$	$\left[-y, \frac{1}{2}, \frac{1}{4}\right]$	`[3,-8]`
4	$[0, -X, Z]$	$\left[y, \frac{1}{2}, \frac{3}{4}\right]$	`[4,-7]`

Table 5: Wyckoff bond: `4b@4b`, bond symmetry: `.2.`

No.	vector	center	mapping
1	$[0, X, 0]$	$\left[\frac{1}{2}, y, 0\right]$	`[1,5]`

continued ...

Table 5

No.	vector	center	mapping
2	$[0, -X, 0]$	$\left[\frac{1}{2}, -y, \frac{1}{2}\right]$	`[2,6]`
3	$[-X, 0, 0]$	$\left[-y, \frac{1}{2}, \frac{1}{4}\right]$	`[3,8]`
4	$[X, 0, 0]$	$\left[y, \frac{1}{2}, \frac{3}{4}\right]$	`[4,7]`

Table 6: Wyckoff bond: `8c@4b`, bond symmetry: `1`

No.	vector	center	mapping
1	$[X, Y, Z]$	$\left[\frac{1}{2}, y, 0\right]$	`[1]`
2	$[-X, -Y, Z]$	$\left[\frac{1}{2}, -y, \frac{1}{2}\right]$	`[2]`
3	$[-Y, X, Z]$	$\left[-y, \frac{1}{2}, \frac{1}{4}\right]$	`[3]`
4	$[Y, -X, Z]$	$\left[y, \frac{1}{2}, \frac{3}{4}\right]$	`[4]`
5	$[-X, Y, -Z]$	$\left[\frac{1}{2}, y, 0\right]$	`[5]`
6	$[X, -Y, -Z]$	$\left[\frac{1}{2}, -y, \frac{1}{2}\right]$	`[6]`
7	$[Y, X, -Z]$	$\left[y, \frac{1}{2}, \frac{3}{4}\right]$	`[7]`
8	$[-Y, -X, -Z]$	$\left[-y, \frac{1}{2}, \frac{1}{4}\right]$	`[8]`

* Wyckoff site: `4c`, site symmetry: `..2`

Table 7: Wyckoff bond: `4a@4c`, bond symmetry: `..2`

No.	vector	center	mapping
1	$[X, -X, Z]$	$\left[x, x, \frac{3}{8}\right]$	`[1,-7]`
2	$[-X, X, Z]$	$\left[-x, -x, \frac{7}{8}\right]$	`[2,-8]`
3	$[X, X, Z]$	$\left[-x, x, \frac{5}{8}\right]$	`[3,-5]`
4	$[-X, -X, Z]$	$\left[x, -x, \frac{1}{8}\right]$	`[4,-6]`

Table 8: Wyckoff bond: `4b@4c`, bond symmetry: `..2`

No.	vector	center	mapping
1	$[X, X, 0]$	$\left[x, x, \frac{3}{8}\right]$	`[1,7]`
2	$[-X, -X, 0]$	$\left[-x, -x, \frac{7}{8}\right]$	`[2,8]`
3	$[-X, X, 0]$	$\left[-x, x, \frac{5}{8}\right]$	`[3,5]`
4	$[X, -X, 0]$	$\left[x, -x, \frac{1}{8}\right]$	`[4,6]`

Table 9: Wyckoff bond: `8c@4c`, bond symmetry: `1`

No.	vector	center	mapping
1	$[X, Y, Z]$	$\left[x, x, \frac{3}{8}\right]$	`[1]`
2	$[-X, -Y, Z]$	$\left[-x, -x, \frac{7}{8}\right]$	`[2]`
3	$[-Y, X, Z]$	$\left[-x, x, \frac{5}{8}\right]$	`[3]`
4	$[Y, -X, Z]$	$\left[x, -x, \frac{1}{8}\right]$	`[4]`
5	$[-X, Y, -Z]$	$\left[-x, x, \frac{5}{8}\right]$	`[5]`
6	$[X, -Y, -Z]$	$\left[x, -x, \frac{1}{8}\right]$	`[6]`

continued ...

Table 9

No.	vector	center	mapping
7	$[Y,\, X,\, -Z]$	$\left[x,\, x,\, \frac{3}{8}\right]$	[7]
8	$[-Y,\, -X,\, -Z]$	$\left[-x,\, -x,\, \frac{7}{8}\right]$	[8]

* Wyckoff site: 8d, site symmetry: 1

Table 10: Wyckoff bond: 8a@8d, bond symmetry: 1

No.	vector	center	mapping
1	$[X,\, Y,\, Z]$	$[x,\, y,\, z]$	[1]
2	$[-X,\, -Y,\, Z]$	$\left[-x,\, -y,\, z+\frac{1}{2}\right]$	[2]
3	$[-Y,\, X,\, Z]$	$\left[-y,\, x,\, z+\frac{1}{4}\right]$	[3]
4	$[Y,\, -X,\, Z]$	$\left[y,\, -x,\, z+\frac{3}{4}\right]$	[4]
5	$[-X,\, Y,\, -Z]$	$[-x,\, y,\, -z]$	[5]
6	$[X,\, -Y,\, -Z]$	$\left[x,\, -y,\, \frac{1}{2}-z\right]$	[6]
7	$[Y,\, X,\, -Z]$	$\left[y,\, x,\, \frac{3}{4}-z\right]$	[7]
8	$[-Y,\, -X,\, -Z]$	$\left[-y,\, -x,\, \frac{1}{4}-z\right]$	[8]